

Sivan Syed



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EDUCATION

The University of Texas at Austin (GPA: 3.8/4.0)

May 2027

Bachelor of Science, **Electrical and Computer Engineering**

Bachelor of Science, **Physics**

Certificate, **German**

Minor, **Materials Science and Engineering**

Relevant Coursework: Fluid Dynamics, Mechanics of Solids, Materials Engineering, Embedded Systems, Electrodynamics

Graduate Coursework: Magnetic Materials and Devices, Computational Physics, Quantum Mechanics

EXPERIENCE

Adom Industries Inc. / R&D Design Engineer / Electrical and Robotics Engineering

May 2025 – August 2025

- Architected and built an **autonomous 48V robot** from the ground up, integrating the mechatronic drivetrain with a custom Vehicle Control Unit (VCU) for **ROS path planning** via **CAN bus**
- Optimized chassis topology through iterative **Finite Element Analysis (FEA)**, reducing overall material usage by 88% while maintaining structural integrity for **reliable payload transfer**
- Authored comprehensive **Design for Manufacturing (DFM)** documentation and bring-up procedures, specifically addressing high-voltage wiring looms, thermal interfaces, and EMI shielding

Sandia National Laboratories / R&D Computational Engineer / Pulsed Power Physics

May 2024 – August 2024

- Built a fast **current-pulse generator** to model rapid capacitor discharges, synthesizing 10k+ waveforms to replicate Z-Machine targeting for **ML-aided surrogate modeling** and **Uncertainty Quantification (UQ)**
- Developed complex **ODE/trajectory solvers** to track radial Beryllium liner implosions, comparing ALEGRA simulation predictions against observed **fluid instability spectra (RTIs/RMIs)**

FSAE Longhorn Racing / Dynamics + Vehicle Modeling Engineer / Steering Lead, Member

September 2022 – August 2025

- Re-engineered the **vehicle steering architecture** from a heavy steel double-u-joint to a **carbon-fiber layup** with a **custom metal 3D-printed bevel gearbox**, reducing system weight by 68% while preserving torsional rigidity
- Defined target vehicle parameters by building a **system-level Model-in-the-Loop (MIL) Python simulation** to optimize anti-Ackermann geometry, **roll center**, and **bump steer** minimization
- Validated tire modeling and slip-angle targets by **tuning compliance coefficients** and correlating simulation outputs against physical **track telemetry**

Projects

Ping Pong Ball Equilibrium Robot

Relevant Skills: *Mechatronics, Robotics, Controls, C, C++, Power Electronics*

- Developing inverse kinematics and PID control** of a **robotic platform** for precise movement correction within ± 0.5 mm accuracy
- Integrating **high-torque servos**, **power circuits**, and custom **3D-printed components** to engineer the **electromechanical system**

Sensor Mounts for Scintillating Fibers

Relevant Skills: *Python, SOLIDWORKS, Manufacturing, Sensor Integration*

- Designed, 3D printed, and milled **custom mechanical mounts** to house a complex sensor array consisting of **SiPMs**, **PMTs**, and **Spectrometers**
- Executed **data analysis pipelines** in **Python** to measure the attenuation lengths of >100 wavelength-shifting fibers using 520nm LEDs

Dance-Dance Revolution Augmented Reality

Relevant Skills: *Computer Vision, OpenCV, HTML, CSS, JavaScript, Three.js*

- Implemented **real-time object detection** via **background subtraction** and **contour detection**, improving accuracy by 89%
- Integrated and synchronized OpenCV movement tracking** with **3D JavaScript graphics** in a **Three.js-rendered** game environment

E-Bike Boost Converter Design

Relevant Skills: *Power Electronics, Embedded Systems, PCB Design, KiCAD, Circuit Design*

- Engineered a custom PCB for a **high-power e-bike DC-DC boost converter**, mitigating radiated EMI and voltage spikes by minimizing the high di/dt **critical loop** and gate-drive parasitic inductance
- Architected a **digital control loop** on a **TI C2000 MCU**, synchronizing the **ePWM module** with the ADC via hardware triggers to sample feedback without switching noise for **precise duty-cycle regulation**

SKILLS

Software Language Proficiencies: Rust, C, C++, ARM, Python, MATLAB, Bash, JS, HTML, CSS, LaTeX

Simulation/Modeling Software: Fusion360, KiCAD, LTSpice, FLASH, MaGe, SOLIDWORKS, VCarve, Inventor, Ansys